

**FEEDBACK
ELECTRONICS**

Deflections detected by the laser are used in a feedback loop to control the position of the tip and keep it close to the sample surface.

PHOTODIODE

A light-sensitive diode tracks the change in height of the tip as it scans across the surface, using the reflected laser beam.

PHOTODIODE

Fine detail
AFM gives scientists a glimpse into the nanoscale world. They build an image from the responses of a probe traveling across a surface.

CANTILEVER

The tip of an AFM probe can be 5 to 40 nanometers in diameter—smaller than most viruses.